

L&T TECHGIUM • PROJECT REGISTRATION ABSTRACT

EconoPulse AI: Autonomous Multi-Modal Global Intelligence & Geospatial Telemetry Platform

Real-Time Threat Monitoring, Macroeconomic Telemetry & 3D Geospatial Infrastructure Discovery

PROJECT TITLE:	EconoPulse AI — Real-Time Global Intelligence & Geospatial Telemetry Platform
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TEAM STRUCTURE:	Individual Participant (Solo Entry)
TECHNICAL TRACK:	Smart Infrastructure, Geospatial Systems & Applied AI

PROJECT ABSTRACT

Exact Count: 164 words

Modern socio-economic disruptions, supply chain shocks, and regional crises unfold rapidly, yet individuals and organizational planners struggle with fragmented data streams and delayed analytical reporting. The aim of this project is to develop EconoPulse AI, an autonomous global intelligence and geospatial telemetry platform that democratizes real-time situational awareness. The proposed solution unites live macro-level event telemetry with an interactive 3D digital-twin globe, enabling users to monitor worldwide crises and seamlessly drill down to micro-level geographic infrastructure. Built using React, Three.js, Vite, and Node.js with Express, the architecture integrates OpenStreetMap, Overpass API, Nominatim spatial geocoding, and Google Gemini LLM reasoning engines. What makes this idea innovative is its dual-layer telemetry synthesis: pairing un-fabricated, ground-truth OpenStreetMap geospatial discovery with automated persona-based impact forecasting that translates global macroeconomic shocks into localized operational guidance. The expected outcome is a unified, high-performance web platform that empowers citizens, logistics operators, and researchers to visualize emerging threats, locate essential regional facilities instantly, and execute timely, data-driven decisions during critical global events.

PROBLEM STATEMENT

TECHNICAL INNOVATION

TOOLS & FRAMEWORKS

Data fragmentation and delayed crisis reporting during global shocks leave individuals and planners without actionable regional intelligence.

Dual-layer telemetry combining live macroeconomic shocks with ground-truth OpenStreetMap spatial discovery and zero synthetic hallucination.

React 19, Three.js 3D Globe, Node.js / Express, Overpass API, Nominatim Geocoding, Google Gemini LLMs, and Vercel Cloud.

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Deployment: <https://global-intel-telemetry.vercel.app>